| | | |
|---|---|---|
| 3(a) | (thermal) energy per unit mass (to change state) | **B1** |
| | change of state without any change of temperature | **B1** |
| 3(b)(i) | 140 g | **A1** |
| 3(b)(ii) | temperature difference (between apparatus and surroundings) does not change | **B1** |
| 3(b)(iii) | $VIt = mL$ | **C1** |
| | $(\{15.1 \times 3.6\} + R) \times 600 = 140 \times L$<br>**or**<br>$(\{7.3 \times 1.8\} + R) \times 600 = 65 \times L$ | **C1** |
| | $41.22 \times 600 = 75 \times L$ | **C1** |
| | $L = 330\ \text{J g}^{-1}$ | **A1** |
| 3(b)(iv) | $15.1 \times 3.6 \times 600 = (140 \times 330) - H$<br>**or**<br>$7.3 \times 1.8 \times 600 = (65 \times 330) - H$ | **C1** |
| | $H = 13\,600$<br><br>rate of gain $= 13\,600 / 600$<br><br>$= 23\ \text{W}$ | **A1** |